

Parametric Equations



Derivatives of parametric equations

- 1 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t .
 - 2 For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.
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Second derivatives of parametric equations

- 3 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\frac{dy}{dx})}{\frac{dx}{dt}}$.
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Tangent lines to parametric curves

- 4 Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.
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Arc length of parametric curves

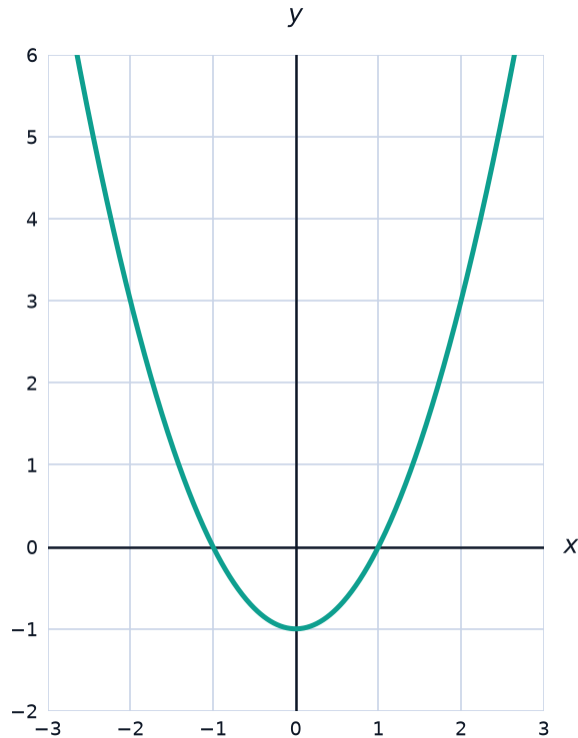
- 5 State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.
 - 6 Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.
 - 7 Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.
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Speed as a vector quantity

- 8 A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.
 - 9 A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t , and interpret the result.
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Eliminating the parameter

- 10 Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.



- 11 Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.
- 12 Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

Concavity of parametric curves

- 13 For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

Horizontal and vertical tangents

- 14 For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).
- 15 For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

Motion along a parametric path (particle interpretation)

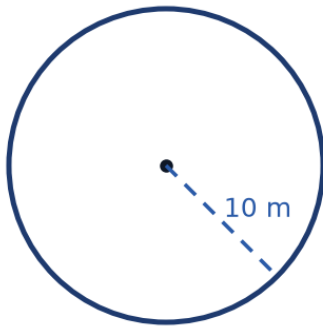
- 16 A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).
- 17 For the particle above, find the speed at $t = 2$.
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Arc length of a full circle (verifying the circumference formula)

- 18 Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r \cos t$, $y(t) = r \sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.
- 19 Find the arc length of the parametric curve $x(t) = e^t \cos t$, $y(t) = e^t \sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.
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Parametric equations from real motion

- 20 A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.



- 21 For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).
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Multiple choice — easy 15

22 For $x(t) = 6t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $-\frac{3}{2}$

b. -6

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

23 For $x(t) = 5t^2$, $y(t) = -5t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{3}{2}$

b. $-\frac{1}{2}$

c. $-\frac{2}{3}$

d. $\frac{3}{2}$

24 For $x(t) = 2t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -4

b. 9

c. -9

d. -8

25 For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 2$.

a. 1

b. 2

c. -1

d. 4

26 For $x(t) = 4t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{9}{4}$

b. -4

c. $-\frac{9}{4}$

d. $-\frac{5}{4}$

27 For $x(t) = t^2$, $y(t) = -6t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. -8

b. -9

c. 9

d. $-\frac{1}{9}$

28 For $x(t) = 4t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{5}{2}$

b. $\frac{3}{2}$

c. $\frac{32}{3}$

d. $-\frac{3}{2}$

29 For $x(t) = 5t^2$, $y(t) = 6t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. $-\frac{54}{5}$

b. $\frac{10}{3}$

c. $\frac{59}{5}$

d. $\frac{54}{5}$

30 For $x(t) = 3t^2$, $y(t) = -t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{3}{2}$

b. $-\frac{1}{2}$

c. -6

d. $-\frac{3}{2}$

31 For $x(t) = 6t^2$, $y(t) = -3t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{4}{3}$

b. $-\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{3}{4}$

32 For $x(t) = 6t^2$, $y(t) = 4t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -6

b. 6

c. 7

d. $\frac{13}{2}$

33 For $x(t) = t^2$, $y(t) = 5t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{8}{15}$

b. -30

c. 31

d. 30

34 For $x(t) = 3t^2$, $y(t) = -6t^3$, find $\frac{dy}{dx}$ at $t = 3$.

- a. 9
- b. -8
- c. -9
- d. -1

35 For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 5$.

- a. $-\frac{5}{2}$
- b. $\frac{7}{2}$
- c. 10
- d. $\frac{5}{2}$

36 For $x(t) = 5t^2$, $y(t) = -4t^3$, find $\frac{dy}{dx}$ at $t = 3$.

- a. $-\frac{13}{5}$
- b. $\frac{18}{5}$
- c. $-\frac{18}{5}$
- d. $-\frac{5}{2}$

Multiple choice — medium 15

37 Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.

- a. $-\frac{7}{2}$
- b. $-\frac{5}{2}$
- c. $-\frac{2}{7}$
- d. $\frac{7}{2}$

- 38 Eliminate the parameter from $x(t) = 6t + 6$, $y(t) = -9t + 8$ to find the Cartesian equation, then state its slope.
- a. $-\frac{3}{2}$
 - b. $-\frac{2}{3}$
 - c. $-\frac{1}{2}$
 - d. $\frac{3}{2}$
- 39 Eliminate the parameter from $x(t) = 4t - 1$, $y(t) = 7t + 6$ to find the Cartesian equation, then state its slope.
- a. $-\frac{7}{4}$
 - b. $\frac{11}{4}$
 - c. $\frac{7}{4}$
 - d. $\frac{4}{7}$
- 40 Eliminate the parameter from $x(t) = 2t - 9$, $y(t) = -t + 4$ to find the Cartesian equation, then state its slope.
- a. $-\frac{1}{2}$
 - b. -2
 - c. $\frac{1}{2}$
 - d. 0
- 41 Eliminate the parameter from $x(t) = t - 6$, $y(t) = -9t + 3$ to find the Cartesian equation, then state its slope.
- a. $-\frac{1}{9}$
 - b. 9
 - c. -8
 - d. -9

- 42 Eliminate the parameter from $x(t) = -5t - 4$, $y(t) = -4t - 7$ to find the Cartesian equation, then state its slope.
- a. $-\frac{4}{5}$
 - b. $\frac{5}{4}$
 - c. $\frac{4}{5}$
 - d. $\frac{9}{5}$
- 43 Eliminate the parameter from $x(t) = 4t - 7$, $y(t) = -5t + 9$ to find the Cartesian equation, then state its slope.
- a. $-\frac{4}{5}$
 - b. $-\frac{5}{4}$
 - c. $-\frac{1}{4}$
 - d. $\frac{5}{4}$
- 44 Eliminate the parameter from $x(t) = t + 1$, $y(t) = 2t + 5$ to find the Cartesian equation, then state its slope.
- a. 3
 - b. -2
 - c. $\frac{1}{2}$
 - d. 2
- 45 Eliminate the parameter from $x(t) = 6t - 5$, $y(t) = -2t + 5$ to find the Cartesian equation, then state its slope.
- a. $\frac{2}{3}$
 - b. $-\frac{1}{3}$
 - c. -3
 - d. $\frac{1}{3}$

- 46 Eliminate the parameter from $x(t) = 5t - 6$, $y(t) = -3t + 5$ to find the Cartesian equation, then state its slope.
- a. $-\frac{5}{3}$
 - b. $\frac{3}{5}$
 - c. $-\frac{3}{5}$
 - d. $\frac{2}{5}$
- 47 Eliminate the parameter from $x(t) = -4t + 3$, $y(t) = -t + 5$ to find the Cartesian equation, then state its slope.
- a. $-\frac{1}{4}$
 - b. $\frac{1}{4}$
 - c. 4
 - d. $\frac{5}{4}$
- 48 Eliminate the parameter from $x(t) = t + 6$, $y(t) = -7t - 4$ to find the Cartesian equation, then state its slope.
- a. -7
 - b. $-\frac{1}{7}$
 - c. 7
 - d. -6
- 49 Eliminate the parameter from $x(t) = 4t + 1$, $y(t) = 9t + 7$ to find the Cartesian equation, then state its slope.
- a. $-\frac{9}{4}$
 - b. $\frac{13}{4}$
 - c. $\frac{9}{4}$
 - d. $\frac{4}{9}$

- 50 Eliminate the parameter from $x(t) = -6t$, $y(t) = -6t + 9$ to find the Cartesian equation, then state its slope.
- a. 1
 - b. 2
 - c. $\frac{3}{2}$
 - d. -1
- 51 Eliminate the parameter from $x(t) = 3t - 1$, $y(t) = 2t + 2$ to find the Cartesian equation, then state its slope.
- a. $-\frac{2}{3}$
 - b. $\frac{2}{3}$
 - c. $\frac{5}{3}$
 - d. $\frac{3}{2}$
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Multiple choice — hard 15

- 52 For $x(t) = 3t^2$, $y(t) = 2t^3 + 6t$, find $\frac{dy}{dx}$ at $t = 4$.
- a. $\frac{21}{4}$
 - b. $\frac{17}{4}$
 - c. $\frac{4}{17}$
 - d. $-\frac{17}{4}$
- 53 For $x(t) = 3t^2 + 5$, $y(t) = t^3 - 4t$, find $\frac{dy}{dx}$ at $t = 1$.
- a. $\frac{1}{6}$
 - b. $\frac{5}{6}$
 - c. -6
 - d. $-\frac{1}{6}$

54 For $x(t) = 5t^2 + 4$, $y(t) = 2t^3 + 5t$, find $\frac{dy}{dx}$ at $t = 4$.

a. $-\frac{101}{40}$

b. $\frac{141}{40}$

c. $\frac{40}{101}$

d. $\frac{101}{40}$

55 For $x(t) = 2t^2 - 6$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{25}{2}$

b. $\frac{23}{2}$

c. $\frac{2}{23}$

d. $-\frac{23}{2}$

56 For $x(t) = 3t^2 - 5$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{26}{3}$

b. $\frac{3}{23}$

c. $-\frac{23}{3}$

d. $\frac{23}{3}$

57 For $x(t) = t^2 + 2t$, $y(t) = t^3 + 6t^2$, find the larger value of t where the tangent line is horizontal.

a. -4

b. 0

c. -3

d. 3

- 58 For $x(t) = t^2$, $y(t) = t^3 - 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.
- a. 2
 - b. 4
 - c. 6
 - d. 8
- 59 For $x(t) = t^2 - 2t$, $y(t) = t^3 + 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.
- a. -2
 - b. -6
 - c. 8
 - d. -4
- 60 For $x(t) = t^2 - 3t$, $y(t) = t^3 + 3t^2 - 24t$, find the larger value of t where the tangent line is horizontal.
- a. -4
 - b. -2
 - c. -8
 - d. 2
- 61 For $x(t) = t^2 + 5t$, $y(t) = t^3 - 27t$, find the larger value of t where the tangent line is horizontal.
- a. 3
 - b. 0
 - c. -9
 - d. -3

- 62 A particle moves with position $x(t) = t^2$, $y(t) = 7t$. Find its speed at $t = 12$.
- a. 7
 - b. 25
 - c. 24
 - d. 27
- 63 A particle moves with position $x(t) = t^2$, $y(t) = 6t$. Find its speed at $t = 4$.
- a. 12
 - b. 8
 - c. 6
 - d. 10
- 64 A particle moves with position $x(t) = t^2$, $y(t) = 3t$. Find its speed at $t = 2$.
- a. 3
 - b. 5
 - c. 7
 - d. 4
- 65 A particle moves with position $x(t) = t^2$, $y(t) = 8t$. Find its speed at $t = 3$.
- a. 6
 - b. 10
 - c. 8
 - d. 12

66 A particle moves with position $x(t) = t^2$, $y(t) = 12t$. Find its speed at $t = 8$.

a. 22

b. 20

c. 16

d. 12